IPL WIN PROBABILITY PREDICTION SYSTEM

Submitted in partial fulfillment of the requirements of

The Degree.

BACHELOR OF ENGINEERING IN

INFORMATION TECHNOLOGY

By

SAHIL KONKAR (16)

SHREYA PATIL (23)

AMIT MISHRA (21)

Supervisor

Prof. Arti Devmane



Department of Information Technology

Shivajirao S. Jondhale College of Engineering., Dombivli (E)

(Affiliated to University of Mumbai)

(AY 2025-26)

CERTIFICATE

This is to certify that the Mini Project entitled **" IPL WIN PROBABILITY PREDICTION SYSTEM "** is a bonafide work of S**ahil Konkar (16) , Shreya Patil (23) , Amit Mishra (21)** submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of "Bachelor of Engineering" in "Information Technology" .

Prof. Arti Devmane
Supervisor

Prof. Arti Devmane
Project Coordinator

Dr. Savita Sangam
Head of Department

Dr. Uttara Gogate
Principal

Internal Examiner

External Examiner

MINI PROJECT APPROVAL

This Mini Project entitled **“ IPL WIN PROBABILITY PREDICTION SYSTEM ”** by **Sahil Konkar(16),Shreya Patil (23) , Amit Mishra (21)** is approved for the degree of Bachelor of Engineering in Information Technology .

Examiners

1……
(InternalExaminerName &Sign)

2……
(External Examiner name & Sign)

Date:

Place:

Contents :

Sr.no	Table Of Content	Pg.no
	Acknowledgement	ii
	Abstract	iii
1.	Introduction	1
1.1	Introduction	2
1.2	Motivation	4
2.	Literature Survey	5
2.1	Survey of Existing System	6
2.2	Limitation Existing system or Research gap	10
3.	Implemented System	11
3.1	Introduction	12
3.2	Architecture / Framework	15
3.3	Algorithm and Process Design	16
3.4	Details of Hardware & Software	17
4.	Result	18
4.1	Result	19
4.2	Conclusion	25
	References	26

FIGURE NO.	LIST OF FIGURES	PAGE NO.
4.1.1	DATA LOADING	20
4.1.2	RAW DATASET 1	20
4.1.3	RAW DATASET 2.1	20
4.1.3	RAW DATASET 2.2	21
4.1.4	DATA PREPROCESSING	22
4.1.5	FEATURE SELECTION	22
4.1.6	MODEL TRAINING	23
4.1.7	MODEL SUMMARY	23
4.1.8	DATA OUTPUT 1	24
4.1.9	DATA OUTPUT 2	24

ABSTRACT :

The project **"IPL Win Probability Prediction System Using Machine Learning"** is a data- driven application designed to predict the probability of a team winning during an ongoing Indian Premier League (IPL) match based on real-time match conditions. The system uses historical IPL datasets obtained from Kaggle, which include match data and ball-by-ball delivery data from 2008 to 2024, and processes them to extract important features such as runs left, balls left, wickets remaining, current run rate (CRR), required run rate (RRR), and city. These features are used as input parameters for the prediction model, which is built using the Light Gradient Boosting Machine (LightGBM) algorithm in Anaconda Jupyter Notebook to ensure high accuracy and efficient performance. The frontend of the system is developed using Streamlit, providing a simple and interactive interface where users can input details like batting team, bowling team, target score, current score, overs left, and wickets left, and receive the winning probability for both teams in percentage form. Overall, the project demonstrates the practical application of machine learning in sports analytics by delivering real-time predictions, improving user engagement, and transforming complex match data into meaningful insights.

ACKNOWLEDGMENTS

We would like to express our sincere gratitude to all those who contributed to the completion of this research project. Special thanks to our supervisor/advisor **Prof. Arti Devmane** for their guidance, support, and valuable insights throughout the development process. We also extend our appreciation to the participants of the user study for their time and feedback, which was instrumental in refining the application's usability. Additionally, we acknowledge the support provided by Shivajirao S. Jondhale College of Engineering for facilitating the resources necessary for this research. Lastly, we are grateful to our families and friends for their encouragement and understanding during the duration of this project.

We would also like to thank our respected Principal **Dr. Uttara Gogate** and our Head of Department **Dr. Savita Sangam** for allowing us to build and present this project.

By:

1. Sahil Konkar (16)
2. Shreya Patil (23)
3. Amit Mishra (21)

CHAPTER 1 : INTRODUCTION

Introduction :

In today's digital era, data analytics and machine learning are widely used in sports like cricket, where the IPL generates large amounts of match data that can be used to predict outcomes more accurately than traditional methods based on basic statistics or expert opinion. The IPL Win Probability Prediction System uses machine learning with features such as runs left, balls left, wickets remaining, CRR, RRR, and city to predict match results in real time. Using historical IPL data and the LightGBM algorithm, the system provides accurate win probabilities and demonstrates the practical application of machine learning in sports analytics. It is implemented using Python with a user-friendly interface and offers dynamic insights during live matches for better analysis and decision-making.

Purpose of the Project :

The IPL Win Probability Prediction System aims to provide:

The main purpose of the IPL Win Probability Prediction System is to provide a reliable and efficient platform that can predict the probability of a team winning during an ongoing IPL match. It aims to transform raw match data into meaningful insights by using machine learning techniques, allowing users to better understand match situations in real time. The system simplifies the process of prediction by analyzing important parameters such as score, overs, wickets, and target, and presenting the results in an easy-to-understand percentage format. Overall, the project focuses on enhancing user experience, improving decision-making, and demonstrating the practical application of data analytics in sports.

Scope :

This project focuses on:

The IPL Win Probability Prediction System covers the complete development of a machine learning-based prediction system using historical IPL datasets from 2008 to 2024. The system includes data collection, preprocessing, feature extraction, model training, and frontend development. Users can input match details such as teams, score, overs, and wickets to receive real-time probability predictions. The project ensures accurate data handling, efficient model performance, and a user-friendly interface. It can be further extended to include live match data integration, advanced visualization, and support for other sports or cricket formats.

FEATURES OF IPL WIN PROBABILITY PREDICTION SYSTEM :

Need for Real-Time Sports Analytics Systems :

With the growing popularity of cricket and the increasing availability of match data, users expect more advanced insights beyond traditional scoreboards. Conventional systems only display basic information such as runs and wickets, which do not fully capture the dynamic nature of the game. Therefore, there is a need for a system that can analyze real-time match conditions and provide meaningful predictions. The IPL Win Probability Prediction System fulfills this requirement by offering probability-based insights, making the match more engaging and informative for users.

Increasing Importance of Data Accuracy and Model Performance :

With the increasing use of machine learning in prediction systems, the accuracy and reliability of the model have become critical factors. The system processes large volumes of IPL data, and any inconsistency or error in the dataset can affect the prediction results. Therefore, proper data preprocessing, cleaning, and feature engineering are essential to ensure high-quality input for the model. Additionally, the use of an efficient algorithm like LightGBM improves prediction accuracy and reduces computation time, ensuring that the system delivers fast and reliable results..

Enhancing User Interaction and Experience :

User interaction plays a vital role in the success of any application. Many analytical systems are complex and difficult for users to understand. The IPL Win Probability Prediction System addresses this challenge by providing a simple and interactive interface using Streamlit. Users can easily input match details and receive predictions instantly without requiring technical knowledge. This improves accessibility and ensures that the system can be used by a wide range of users, including cricket fans and analysts.

Promoting Accessibility and User-Friendly Design :

In today’s digital age, applications must be designed to be accessible and easy to use across different devices. The IPL prediction system is developed with a user-friendly interface that ensures smooth navigation and clear presentation of results. It is compatible with desktops, tablets, and mobile devices, allowing users to access it anytime during a match.

1.2 MOTIVATION FOR PROJECT :

The primary motivation behind developing the IPL Win Probability Prediction System is the increasing demand for real-time sports analytics and the growing interest of users in understanding match outcomes beyond basic scoreboards. In today's fast-paced environment, cricket fans prefer instant insights that can help them analyze match situations dynamically rather than relying only on traditional statistics. Existing systems do not provide an easy way for users to interact with data and generate predictions based on current match conditions. This project aims to bridge that gap by providing a system where users can input match details and instantly receive probability-based results.

Another motivating factor is the opportunity to apply machine learning concepts to a real-world problem using actual IPL datasets. Working with large-scale data helps in understanding data preprocessing, feature engineering, and model training in a practical manner. It also enhances technical knowledge by integrating tools such as Python, LightGBM, and Streamlit into a single system. This project not only improves problem-solving skills but also demonstrates how data science can be used effectively in sports analytics.

Finally, the project is motivated by the need to combine advanced technology with user-friendly design. Many existing analytical systems are complex and not easily accessible to general users. By developing a simple and interactive interface, this system ensures that even non-technical users can benefit from machine learning predictions. Overall, the project aims to create a balance between technical accuracy and usability, delivering a system that is efficient, reliable, and easy to use.

CHAPTER 2 : LITERATURE SURVEY

Literature Survey :

[1] In 2022, a study by Saurabh Kumar published in the Journal of Prediction Markets focused on predicting the outcome of IPL cricket matches using machine learning techniques. The system used features such as player ranking, auction price, historical match data, and team strength to analyze match results and identify important factors influencing outcomes.

Advantages: Identifies key factors affecting IPL match results and uses multiple parameters for better analysis

Limitations: Limited dataset used and does not include live match features.

[2] In 2023, Sangaboina and Kumar presented their research in the International Journal of Engineering and Computer Science on win probability prediction for IPL matches using machine learning techniques. The model considered features such as target score, run rate, wickets fallen, and score progression, and applied the Random Forest algorithm for prediction.

Advantages: Provides high prediction accuracy using Random Forest and considers match progression factors.

Limitations: Requires real-time data for better accuracy and may suffer from overfitting.

[3] In 2022, Sneha Singh and her team published a study in the Journal of Machine Learning and Cloud Computing focusing on cricket match prediction using machine learning. The system used parameters such as toss result, venue, weather conditions, and team strength, and applied multiple algorithms for analysis.

Advantages: Uses multiple machine learning algorithms and provides detailed analysis.

Limitations: Uses general cricket data instead of IPL-specific and lacks focus on real-time prediction.

[4] In 2025, Zohaib et al. proposed an IPL win probability prediction system in the International Journal for Research in Applied Science and Engineering Technology. The system used historical IPL data, team statistics, and player performance data, and applied Logistic Regression and Gradient Boosting algorithms.

Advantages: Uses Logistic Regression and Gradient Boosting and provides structured prediction model.

Limitations: Depends heavily on data quality and has limited real-time adaptability.

[5] In 2025, Mahendra Labana and his team presented a study in the International Journal of Advanced Research in Computer and Communication Engineering on IPL team winning prediction using machine learning. The system used features such as player performance, venue, and toss outcome and applied the XGBoost algorithm.

Advantages: Improves prediction accuracy using XGBoost and considers multiple influencing factors.

Limitations: Cricket outcomes are highly unpredictable and model performance varies with data.

[6] In 2020, Srikanth K C and his team published a research paper on arXiv focusing on IPL match outcome prediction using machine learning techniques. The system used features such as player averages, team composition, and venue, and applied the Random Forest algorithm.

Advantages: Random Forest provides good accuracy and works well with structured datasets.

Limitations: Requires large dataset and extensive preprocessing.

2.2 Limitations of the Existing System or Research Gap :

1. **High Computational Requirements:** Some advanced systems require powerful hardware and resources, making them inefficient for regular users.

2. **Limited Accessibility:** Advanced analytics platforms are often restricted or paid, making them inaccessible to general users and students.

3. **No Interactive Frontend:** Most machine learning models available online do not provide a simple and user-friendly interface for real-time prediction.

4. **Absence of Deployment:** Many models remain at experimental stages and are not deployed as usable applications.

5. **Lack of Real-Time Interaction:** Most existing systems do not allow users to input live match conditions and receive instant probability predictions, limiting user engagement.

6. **Complex User Interface:** Many existing systems have complicated dashboards and visualizations, which are difficult for non-technical users to understand.

7. **Dependency on Static Data:** Many systems are based on historical data only and do not adapt well to changing match conditions in real time.

8. **Limited Feature Selection:** Existing models often use fewer parameters, which may reduce prediction accuracy and reliability.

9. **Accuracy Limitations:** Some prediction systems do not achieve high accuracy due to improper data preprocessing and model selection.

10. **Lack of Customization:** Users cannot modify inputs freely to analyze different match scenarios and outcomes.

CHAPTER 3 : IMPLEMENTED SYSTEM

IMPLEMENTED SYSTEM :

The proposed system, IPL Win Probability Prediction System, is a machine learning-based application designed to simplify and automate the process of predicting match outcomes during Indian Premier League (IPL) matches. Users can enter match details such as teams, score, overs, wickets, and target, and obtain real-time win probability predictions, while the system processes data and generates results efficiently. Built with Python for backend processing and Streamlit for the front end, along with LightGBM for prediction, the system provides a responsive, accurate, and user-friendly interface, improving prediction efficiency, reducing manual analysis, and enhancing the overall understanding of match situations.

1. Objectives of the Implemented System :

1. To develop a machine learning-based system for predicting IPL match outcomes based on real-time conditions.

2. To provide a user-friendly interface that allows users to input match details and receive instant predictions.

3. To process and analyze historical IPL datasets for accurate model training.

4. To implement an efficient algorithm such as LightGBM for better performance and accuracy.

5. To demonstrate the practical application of data science and machine learning in sports analytics.

2. Features of the Our System :

1. **Historical IPL Data:** The system is trained on historical IPL match data, which includes past match outcomes, team performance, and match conditions to build an accurate prediction model.

2. M**achine Learning Model (LightGBM):** The system uses the LightGBM algorithm trained on historical IPL data to provide fast, efficient, and highly accurate match outcome predictions.

3. U**ser Input System:** Users can enter match details such as batting team, bowling team, city, target score, current score, overs left, and wickets left to generate predictions.

4. **Real-Time Prediction:** The system instantly calculates and displays the winning probability of both teams in percentage format based on the given match inputs.

5. I**nteractive Interface:** A simple and user-friendly interface is developed using Streamlit, allowing users to easily input data and view results.

6. D**ata Processing:** The system performs data cleaning, preprocessing, and feature extraction to ensure high-quality input for the model.

7. A**ccurate Prediction System:** The model is trained on IPL datasets from 2008 to 2024, improving the accuracy and reliability of predictions.

8. P**erformance Optimization:** The use of LightGBM ensures faster computation and quick response time for real-time predictions.

9. S**calability:** The system can be enhanced in the future by adding live data integration, advanced analytics, and visualization features.

10. **Responsive Design:** The application is accessible on different devices such as desktops, tablets, and smartphones without any issues.

11. U**ser-Friendly Experience:** The system provides a smooth and easy experience for users with clear outputs and simple navigation.

Key Components of the Our System :

1. **Accurate Predictions:** The system uses the LightGBM machine learning algorithm trained on historical IPL data, ensuring reliable and data-driven prediction results.

2. **Real-Time Insights:** Users can get instant win probability predictions during live matches, helping them better understand match situations.

3. **User-Friendly Interface:** The Streamlit-based interface is simple and easy to use, allowing even non-technical users to interact with the system.

4. T**ime Efficient:** The system provides quick results without manual calculations, saving time and effort for users.

5. P**ractical Application of ML:** Demonstrates real-world use of machine learning and data analytics in sports prediction, making it useful for learning and analysis.

4.Performance Analytics & Insights :

The IPL Win Probability Prediction System ensures strong performance through the use of an efficient machine learning model and proper data handling techniques. The LightGBM algorithm plays a key role in achieving high prediction accuracy by learning from historical IPL datasets and important match features. The system is capable of processing user inputs quickly and generating predictions in real time, which makes it highly effective during live matches. Data preprocessing and feature engineering further improve model performance by ensuring that only relevant and high-quality data is used. The system provides consistent and reliable outputs across different match scenarios, maintaining stability and accuracy. Additionally, the model is scalable and can handle large datasets, allowing future improvements by incorporating more data and advanced features.

5.User Support :

The IPL Win Probability Prediction System is designed to provide simple and effective user support for smooth interaction. It offers a user-friendly interface where users can easily input match details and obtain predictions without any technical difficulty. Clear input fields and easy navigation help users quickly understand the system. Although advanced support features like live chat are not included, the system ensures reliability through its simple design and fast response. Future enhancements may include help sections and feedback options to further improve user experience.

3.2 System Architecture / Framework :

The IPL Win Probability Prediction System follows a structured and modular architecture that integrates data processing, machine learning, and user interaction components to provide real-time predictions. The system begins with the data layer, where historical IPL datasets, including match data and ball-by-ball delivery data from 2008 to 2024, are collected and stored. This data is then passed to the preprocessing module, where it undergoes cleaning, transformation, and feature engineering to extract important parameters such as runs left, balls left, wickets remaining, current run rate (CRR), and required run rate (RRR).

The processed data is used to train the machine learning model using the LightGBM algorithm in the Anaconda Jupyter Notebook environment. The trained model is then saved and integrated into the application for real-time prediction. When a user interacts with the system through the frontend interface developed using Streamlit, the input data is sent to the backend, where the trained model processes the inputs and generates the winning probability for both teams. The output is then displayed to the user in percentage format through the interface.

Overall, the system architecture ensures smooth data flow between components, efficient processing, and quick response time. It follows a pipeline approach consisting of data collection, preprocessing, model training, prediction, and result visualization. This architecture provides scalability, flexibility, and reliability, allowing future enhancements such as live data integration and advanced analytics features.

3.3 Algorithm and Process Design :

The IPL Win Probability Prediction System is based on a structured process that involves data collection, preprocessing, model training, and real-time prediction using a machine learning algorithm. The system primarily uses the Light Gradient Boosting Machine (LightGBM) algorithm, which is known for its high accuracy and fast performance in handling large datasets. Initially, historical IPL datasets, including match data and ball-by-ball delivery data, are collected from reliable sources such as Kaggle. This data is then preprocessed to remove inconsistencies, handle missing values, and convert it into a structured format suitable for analysis.

During the preprocessing stage, important features such as runs left, balls left, wickets remaining, current run rate (CRR), and required run rate (RRR) are extracted from the raw data. These features play a crucial role in determining the outcome of the match and are used as input variables for the model. The cleaned and processed data is then used to train the LightGBM model in the Anaconda Jupyter Notebook environment. The model learns patterns and relationships between input features and match outcomes, enabling it to make accurate predictions.

Once the model is trained, it is integrated into the system for real-time prediction. When a user enters match details through the Streamlit interface, the input data is processed and passed to the trained model. The model then calculates the probability of winning for both teams based on the given conditions and returns the output in percentage format. The entire process is designed to be efficient, accurate, and fast, ensuring smooth user interaction and reliable prediction results. Overall, the algorithm and process design provide a complete pipeline from data handling to prediction and output visualization.

3.4 Details of Software Used:

The development of the **IPL Win Probability Prediction System** involves the use of various software tools and technologies that work together to ensure accurate prediction, efficient processing, and a smooth user experience. The system is primarily developed using Python, which serves as the core programming language due to its simplicity and extensive support for data science and machine learning libraries. For building the prediction model, the Light Gradient Boosting Machine (LightGBM) algorithm is used, as it provides high accuracy, faster training speed, and efficient handling of large datasets.

For data manipulation and preprocessing, libraries such as Pandas and NumPy are used to handle structured IPL datasets effectively. These libraries help in cleaning the data, handling missing values, and transforming raw data into meaningful features required for model training. Visualization tools like Matplotlib and Seaborn are used during the development phase to analyze data patterns and understand relationships between different features, which improves model performance.

The machine learning model is developed and tested using Anaconda Jupyter Notebook, which provides an interactive environment for coding, data analysis, and visualization. It allows step- by-step execution and easy debugging, making it suitable for developing machine learning applications. For building the frontend, Streamlit is used, which is a Python-based framework that enables quick development of web applications. It provides an interactive and user-friendly interface where users can input match details and view prediction results in real time.

Additionally, Visual Studio Code is used as the code editor for writing and managing the project files due to its flexibility and support for multiple extensions. Web browsers such as Google Chrome are used for testing and running the application to ensure proper functionality and responsiveness. Overall, the combination of these software tools ensures that the system is efficient, accurate, and easy to use, providing a complete end-to-end solution for IPL win probability prediction.

CHAPTER 4 : RESULTS

RESULTS :

The **IPL Win Probability Prediction System** successfully provides accurate and real-time predictions based on user inputs and historical IPL data. The system allows users to enter match details such as batting team, bowling team, target score, current score, overs left, and wickets left. Based on these inputs, the system processes the data using the trained LightGBM model and generates winning probabilities for both teams in percentage form. This helps users understand the match situation clearly and effectively.

From the user perspective, the Streamlit-based interface ensures a smooth and interactive experience. Users can easily navigate the application, input the required details, and obtain results instantly without any complexity. The system is designed to be simple and responsive, making it accessible to both technical and non-technical users. The quick response time and clear output presentation enhance usability and overall user satisfaction.

In terms of performance, the system operates efficiently with minimal delay, making it suitable for real-time usage during live matches. The machine learning model ensures reliable and consistent predictions across different match scenarios. The results demonstrate that the system successfully achieves its objectives by providing accurate predictions, improving match understanding, and showcasing the practical application of machine learning in sports analytics.

```
[1]: import numpy as np
     import pandas as pd

[2]: match = pd.read_csv('matches.csv')
     delivery = pd.read_csv('deliveries.csv')
```

FIG NO. 4.1.1 (DATA LOADING)

```
[3]: match.head()
```

[3]:

	id	Season	city	date	team1	team2	toss_winner	toss_decision	result	dl_applied	winner	win_by_runs	win_by_wickets	player_of_match
0	1	IPL-2017	Hyderabad	05-04-2017	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field	normal	0	Sunrisers Hyderabad	35	0	Yuvraj Singh
1	2	IPL-2017	Pune	06-04-2017	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	normal	0	Rising Pune Supergiant	0	7	SPD Smith
2	3	IPL-2017	Rajkot	07-04-2017	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field	normal	0	Kolkata Knight Riders	0	10	CA Lynn
3	4	IPL-2017	Indore	08-04-2017	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	normal	0	Kings XI Punjab	0	6	GJ Maxwell
4	5	IPL-2017	Bangalore	08-04-2017	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat	normal	0	Royal Challengers Bangalore	15	0	KM Jadhav

FIG NO. 4.1.2 (RAW DATASET 1)

```
[5]: delivery.head()
```

[5]:

	match_id	inning	batting_team	bowling_team	over	ball	batsman	non_striker	bowler	is_super_over	...	bye_runs	legbye_runs	noball_runs	penalty_runs	batsr
0	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	1	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	
1	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	2	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	
2	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	3	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	
3	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	4	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	
4	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	5	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	

FIG NO. 4.1.3 (RAW DATASET 2.1)

```
[5]: delivery.head()
```

[5]:

	match_id	inning	batting_team	bowling_team	over	ball	batsman	non_striker	bowler	is_super_over	...	bye_runs	legbye_runs	noball_runs	penalty_runs	batsr
0	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	1	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	
1	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	2	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	
2	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	3	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	
3	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	4	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	
4	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	5	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	

FIG NO. 4.1.3 (RAW DATASET 2.2)

```
[23]: delivery_df['current_score'] = delivery_df.groupby('match_id')['total_runs_y'].cumsum()
[24]: delivery_df['runs_left'] = delivery_df['total_runs_x'] + 1 - delivery_df['current_score']
[25]: delivery_df['target'] = delivery_df['total_runs_x'] + 1
[26]: delivery_df
```

[26]:

bowling_team	over	ball	batsman	...	penalty_runs	batsman_runs	extra_runs	total_runs_y	player_dismissed	dismissal_kind	fielder	current_score	runs_left	target
Sunrisers Hyderabad	1	1	CH Gayle	...	0	1	0	1	NaN	NaN	NaN	1	207	208
Sunrisers Hyderabad	1	2	Mandeep Singh	...	0	0	0	0	NaN	NaN	NaN	1	207	208
Sunrisers Hyderabad	1	3	Mandeep Singh	...	0	0	0	0	NaN	NaN	NaN	1	207	208
Sunrisers Hyderabad	1	4	Mandeep Singh	...	0	2	0	2	NaN	NaN	NaN	3	205	208
Sunrisers Hyderabad	1	5	Mandeep Singh	...	0	4	0	4	NaN	NaN	NaN	7	201	208
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Mumbai Indians	20	2	RA Jadeja	...	0	1	0	1	NaN	NaN	NaN	152	1	153

```
[29]: delivery_df['player_dismissed'] = delivery_df['player_dismissed'].fillna("0")
delivery_df['player_dismissed'] = delivery_df['player_dismissed'].apply(lambda x: 0 if x == "0" else "1")
delivery_df['player_dismissed'] = delivery_df['player_dismissed'].astype('int')
wickets = delivery_df.groupby('match_id')['player_dismissed'].cumsum().values
delivery_df['wickets'] = 10 - wickets
delivery_df.head()
```

[29]:

ng_team	bowling_team	over	ball	batsman	...	extra_runs	total_runs_y	player_dismissed	dismissal_kind	fielder	current_score	runs_left	target	balls_left	wickets
Royal allengers engaluru	Sunrisers Hyderabad	1	1	CH Gayle	...	0	1	0	NaN	NaN	1	207	208	119	10
Royal allengers engaluru	Sunrisers Hyderabad	1	2	Mandeep Singh	...	0	0	0	NaN	NaN	1	207	208	118	10
Royal allengers engaluru	Sunrisers Hyderabad	1	3	Mandeep Singh	...	0	0	0	NaN	NaN	1	207	208	117	10
Royal allengers engaluru	Sunrisers Hyderabad	1	4	Mandeep Singh	...	0	2	0	NaN	NaN	3	205	208	116	10
Royal allengers engaluru	Sunrisers Hyderabad	1	5	Mandeep Singh	...	0	4	0	NaN	NaN	7	201	208	115	10

```
[31]: #crr = runs/overs
delivery_df['crr'] = (delivery_df['current_score']*6)/(120 - delivery_df['balls_left'])

[33]: #rrr = runs left/ overs left
delivery_df['rrr'] = (delivery_df['runs_left']*6)/delivery_df['balls_left']

[35]: def result(row):
          return 1 if row['batting_team'] == row['winner'] else 0

[36]: delivery_df['result'] = delivery_df.apply(result,axis=1)

[37]: delivery_df
```

[37]:

inning	batting_team	bowling_team	over	ball	batsman	...	dismissal_kind	fielder	current_score	runs_left	target	balls_left	wickets	crr	rrr	result
2	Royal Challengers Bengaluru	Sunrisers Hyderabad	1	1	CH Gayle	...	NaN	NaN	1	207	208	119	10	6.000000	10.436975	0
2	Royal Challengers Bengaluru	Sunrisers Hyderabad	1	2	Mandeep Singh	...	NaN	NaN	1	207	208	118	10	3.000000	10.525424	0
2	Royal Challengers Bengaluru	Sunrisers Hyderabad	1	3	Mandeep Singh	...	NaN	NaN	1	207	208	117	10	2.000000	10.615385	0
2	Royal Challengers Bengaluru	Sunrisers Hyderabad	1	4	Mandeep Singh	...	NaN	NaN	3	205	208	116	10	4.500000	10.603448	0
2	Royal Challengers Bengaluru	Sunrisers Hyderabad	1	5	Mandeep Singh	...	NaN	NaN	7	201	208	115	10	8.400000	10.486957	0

FIG NO. 4.1.4 (DATA PREPROCESSING)

```
[39]: final_df = delivery_df[['batting_team','bowling_team','city','runs_left','balls_left','wickets','target','crr','rrr','result']]

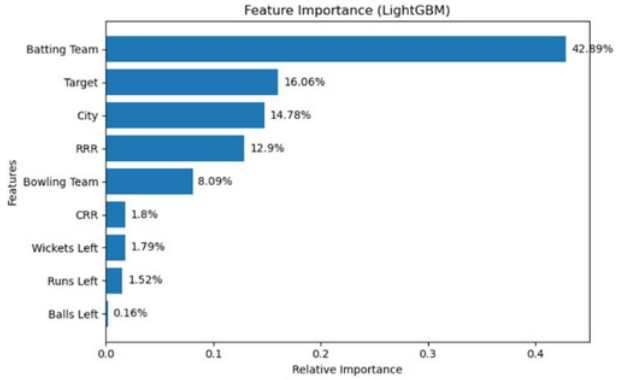
[40]: final_df
```

[40]:

	batting_team	bowling_team	city	runs_left	balls_left	wickets	target	crr	rrr	result
125	Royal Challengers Bengaluru	Sunrisers Hyderabad	Hyderabad	207	119	10	208	6.000000	10.436975	0
126	Royal Challengers Bengaluru	Sunrisers Hyderabad	Hyderabad	207	118	10	208	3.000000	10.525424	0
127	Royal Challengers Bengaluru	Sunrisers Hyderabad	Hyderabad	207	117	10	208	2.000000	10.615385	0
128	Royal Challengers Bengaluru	Sunrisers Hyderabad	Hyderabad	205	116	10	208	4.500000	10.603448	0
129	Royal Challengers Bengaluru	Sunrisers Hyderabad	Hyderabad	201	115	10	208	8.400000	10.486957	0
...	...	...	...	...	...	...	...	...	...	...
159547	Chennai Super Kings	Mumbai Indians	Hyderabad	1	4	5	153	7.862069	1.500000	0
159548	Chennai Super Kings	Mumbai Indians	Hyderabad	-1	3	5	153	7.897436	-2.000000	0
159549	Chennai Super Kings	Mumbai Indians	Hyderabad	-2	2	4	153	7.881356	-6.000000	0
159550	Chennai Super Kings	Mumbai Indians	Hyderabad	-4	1	4	153	7.915966	-24.000000	0
159551	Chennai Super Kings	Mumbai Indians	Hyderabad	-4	0	3	153	7.850000	-inf	0

77212 rows × 10 columns

FIG NO. 4.1.5 (FEATURE SELECTION)



```
[48]: x = final_df.iloc[:,:-1]
      y = final_df.iloc[:,-1]
      from sklearn.model_selection import train_test_split
      x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_state=1)

[49]: x_train
```

[49]:

	batting_team	bowling_team	city	runs_left	balls_left	wickets	target	crr	rrr
85136	Kolkata Knight Riders	Punjab Kings	Kolkata	140	111	8	150	6.666667	7.567568
58683	Sunrisers Hyderabad	Mumbai Indians	Hyderabad	148	101	9	173	7.894737	8.792079
31197	Rajasthan Royals	Chennai Super Kings	Centurion	71	32	6	165	6.409091	13.312500
125226	Kolkata Knight Riders	Delhi Capitals	Delhi	29	10	2	187	8.618182	17.400000
82322	Kolkata Knight Riders	Punjab Kings	Chandigarh	51	35	7	158	7.552941	8.742857
...	...	...	...	...	...	...	...	...	...
43060	Kolkata Knight Riders	Rajasthan Royals	Ahmedabad	101	56	8	169	6.375000	10.821429
44778	Mumbai Indians	Chennai Super Kings	Mumbai	99	71	10	181	10.040816	8.366197
153879	Rajasthan Royals	Mumbai Indians	Jaipur	127	97	9	172	11.739130	7.855670
105507	Rajasthan Royals	Punjab Kings	Chandigarh	97	42	6	180	6.384615	13.857143
11038	Sunrisers Hyderabad	Mumbai Indians	Hyderabad	81	74	9	139	7.565217	6.567568

60900 rows × 9 columns

```
[49]: from sklearn.compose import ColumnTransformer
      from sklearn.preprocessing import OneHotEncoder

      trf = ColumnTransformer([
          ('trf',OneHotEncoder(sparse_output=False,drop='first'),['batting_team','bowling_team','city'])
      ]
      ,remainder='passthrough')

[53]: pipe = Pipeline(steps=[
          ('step1',trf),
          ('step2',LGBMClassifier(
              n_estimators=100,
              learning_rate=0.1,
              random_state=1,
              importance_type='gain'
          ))
      ])

[54]: model = CalibratedClassifierCV(pipe,method='sigmoid',cv=5)

[56]: y_pred = model.predict(x_test)

[57]: from sklearn.metrics import accuracy_score
      accuracy_score(y_test,y_pred)

[57]: 0.9953369236831735

[58]: model.predict_proba(x_test)[88]

[58]: array([9.99809620e-01, 1.90379824e-04])
```

FIG NO. 4.1.6 (MODEL TRAINING)

```
[73]: print("Model Summary:")
      print("--------------")
      print("Algorithm: LightGBM Classifier")
      print("Number of Trees:", lgbm_model.n_estimators)
      print("Learning Rate:", lgbm_model.learning_rate)
      print("Max Depth:", lgbm_model.max_depth)
      print("Num Leaves:", lgbm_model.num_leaves)
      print("Number of Features:", lgbm_model.n_features_in_)
      print("Classes:", lgbm_model.classes_)

      Model Summary:
      --------------
      Algorithm: LightGBM Classifier
      Number of Trees: 100
      Learning Rate: 0.1
      Max Depth: -1
      Num Leaves: 31
      Number of Features: 54
      Classes: [0 1]
```

FIG NO. 4.1.7 (MODEL SUMMARY)

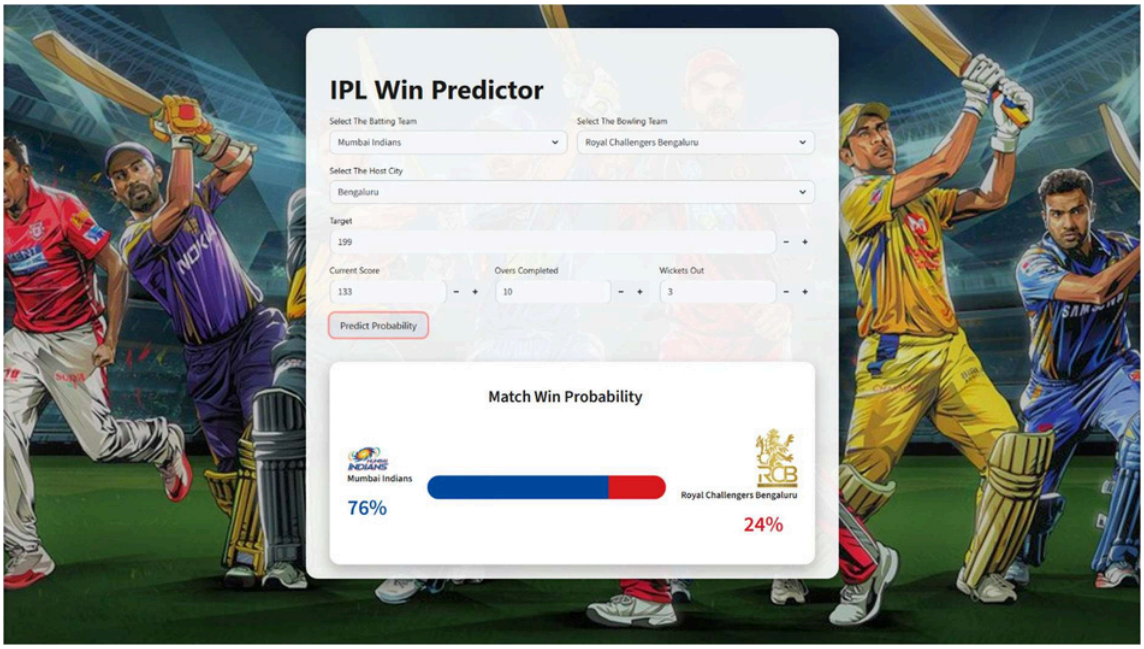


FIG NO. 4.1.8 (DATA OUTPUT)

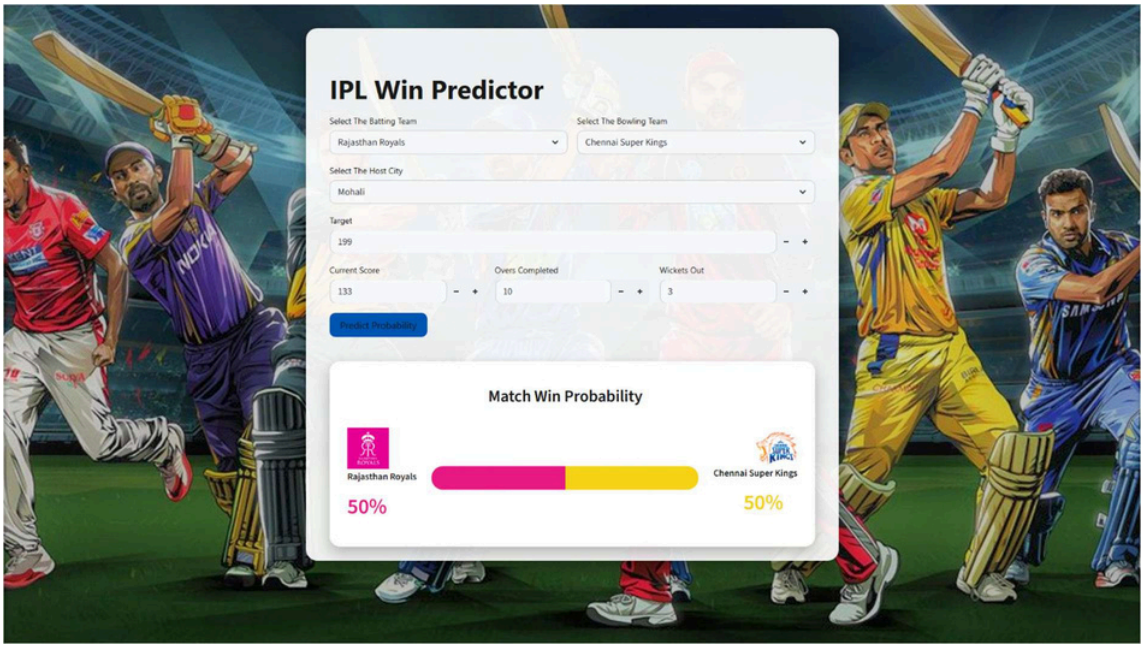


FIG NO. 4.1.9 (DATA OUTPUT)

CONCLUSION :

The IPL Win Probability Prediction System successfully achieves its goal of developing a machine learning-based application for predicting match outcomes in real time. The system uses historical IPL data along with important match parameters such as score, overs, wickets, and target to generate accurate win probability predictions. The implementation of the LightGBM algorithm ensures efficient performance and reliable results, making the system effective for practical use.

From the user perspective, the system provides a simple and interactive interface using Streamlit, allowing users to easily input match details and obtain instant predictions. The design focuses on ease of use, ensuring that both technical and non-technical users can understand and utilize the system without difficulty. The clear presentation of results in percentage form enhances user understanding of match situations and improves overall engagement.

Overall, the project demonstrates the successful integration of machine learning, data analysis, and frontend development into a single application. It highlights the practical application of data science in sports analytics and provides valuable insights into match dynamics. The system serves as a reliable and efficient tool for prediction and also offers scope for future improvements such as live data integration and advanced analytics features.

REFERENCES

1. https://www.kaggle.com/datasets/ramjidoolla/ipl-data-set
2. https://www.kaggle.com/datasets/patrickb1912/ipl-complete-dataset-20082020
3. https://www.computersciencejournals.com/ijecs/archives/2023/vol5issue2/PartA/5-2-3-924.pdf
4. https://www.ijraset.com/best-journal/ipl-win-probability-prediction-system-using-machine-learning-techniques
5. https://matjournals.co.in/index.php/RRMLCC/article/view/349?utm_source=chatgpt.com
6. https://ijarcce.com/wp-content/uploads/2025/11/IJARCCE.2025.141139-ipl.pdf
7. https://arxiv.org/pdf/2110.01395
8. https://www.atlantis-press.com/proceedings/iciic-21/125960812
9. https://arxiv.org/pdf/1809.09813
10. https://learnwith.campusx.in/